

A CMS Open Data Diagnostic Search for $H \rightarrow \tau\tau$ in the $\mu\tau_h$ Final State

Analysis my_analysis¹

¹CMS Open Data reduced-sample analysis
(Dated: June 2, 2026)

A diagnostic search for Higgs boson decays to tau pairs is performed with CMS 2012 Open Data reduced samples in the $\mu\tau_h$ final state. A Phase 5 audit found that an initially selected BDT-score observed fit was not a valid evidence result because the score shape failed data validation. The primary result is therefore a conservative visible-mass fit in VBF, boosted, and zero-jet categories with a same-sign data-driven QCD/fake template. The fit gives $\hat{\mu} = 0.438$, an observed 95% CLs upper limit $\mu < 10.764$, and a discovery diagnostic $Z = 0.095$. The BDT-score fit gives $\hat{\mu} = 9.352$ and $\mu < 14.837$ but is retained only as a flagged diagnostic, not as CMS-quality evidence.

Data and method. The analysis uses the public Run2012B/C TauPlusX HiggsTauTauReduced mirror and the CMS Open Data luminosity reference of 11.467 fb⁻¹. Signal and background simulations are normalized with official Open Data record event denominators rather than local reduced-file entries. Events are selected with the $\mu + \tau_h$ trigger, tight muon and hadronic-tau requirements including a tight anti-muon veto, opposite-sign charge, and low transverse mass.

Background correction. The W+jets normalization is measured in a high- m_T control region. The VBF-background rate is calibrated in a VBF-like top-btag control region outside the low- m_T signal region, giving a scale 0.557 ± 0.051 . The reducible QCD/fake contribution is estimated from same-sign low- m_T data after subtracting non-QCD simulation. For the primary visible-mass fit, the OS/SS transfer factor measured in the lowest visible-mass bin is 0.996 ± 0.120 .

Results. The primary corrected workspace has median expected 95% CLs limit $\mu < 10.738$ and observed limit $\mu < 10.764$. The best fit is $\hat{\mu} = 0.438$ with $Z = 0.095$. This is the conservative final observed result. The categorized BDT-score fit is shown only as a diagnostic because high-score residuals remain after the QCD/fake correction.

MVA diagnostics. Histogram gradient boosting, an MLP neural network, and XGBoost were trained on selected simulation. Their ROC curves and feature-importance diagnostics are retained to document the sensitivity study. A transformer and the requested GenMET regression are downscoped because the fast attention stack and GenMET target are absent from the current reduced workflow.

TABLE I. Selected yields in the primary visible-mass fit after the QCD/fake audit correction. Background includes DY, ttbar, W+jets, and the same-sign QCD/fake estimate.

Category	Data	Bkg.	QCD/fake	Signal
vbf	78	96.3	17.7	1.59
boosted	2183	2364.4	325.6	9.17
zero_jet	8451	8456.8	1569.6	14.34

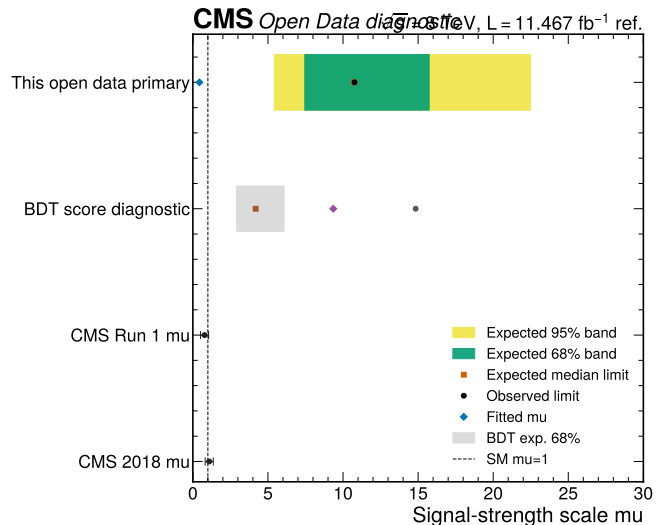


FIG. 1. Signal-strength and limit comparison. The primary open-data result is shown with expected bands and an observed marker following CMS-style limit plot conventions. The BDT score row is flagged and the CMS publication rows are context, not direct validation targets.

rent reduced workflow.

Comparison and interpretation. CMS Run 1 reported evidence for $H \rightarrow \tau\tau$ with a best-fit signal strength of 0.78 ± 0.27 , and CMS later observed the decay at 13 TeV with observed and expected significances of 4.9 and 4.7. The present reduced open-data result is not a reproduction of those analyses: it uses fewer channels, a reduced public sample, limited calibrations, and a simplified background program. Its value is methodological reproducibility, not an independent CMS-quality evidence claim.

[1] CMS Collaboration, Evidence for the 125 GeV Higgs boson decaying to a pair of tau leptons, JHEP 05 (2014) 104.

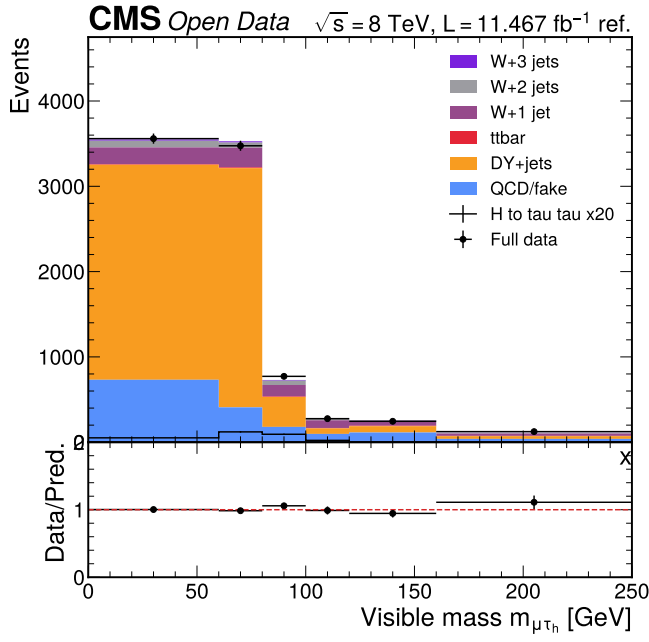


FIG. 2. Primary visible-mass validation in the zero-jet category. The same-sign QCD/fake estimate stabilizes the dominant zero-jet normalization, which was the main pathology in the original score-only observed result.

- [2] CMS Collaboration, Observation of the SM scalar boson decaying to a pair of tau leptons, Phys. Lett. B 779 (2018) 283.
- [3] CERN Open Data Portal, CMS Higgs to tau tau reduced samples for education and outreach.
- [4] L. Heinrich et al., pyhf: pure-Python implementation of HistFactory statistical models.
- [5] A. L. Read, Presentation of search results: the CLs technique.
- [6] G. Cowan et al., Asymptotic formulae for likelihood-based tests of new physics.

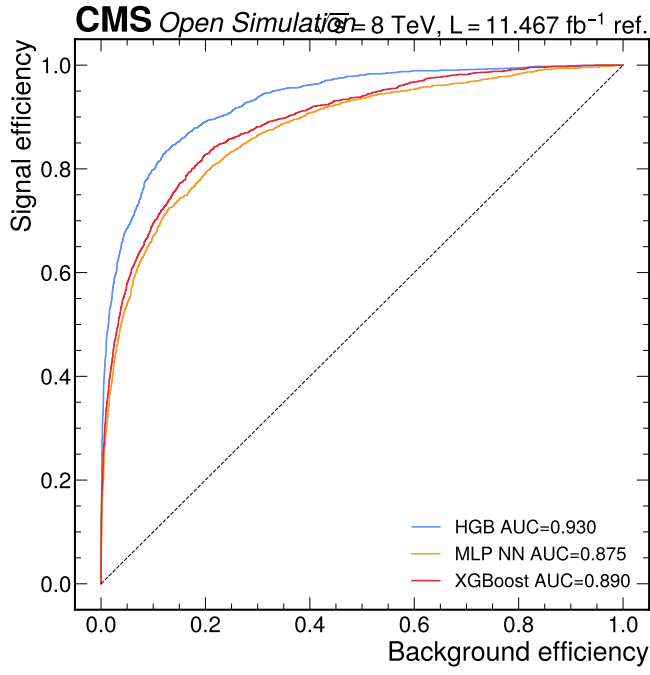


FIG. 3. Classifier ROC curves for the HGB, MLP neural-network, and XGBoost score columns on selected signal and background simulation. These curves show MC separation power but do not validate observed score shapes.